

# JAVA PROGRAMMING LABORATORY

Experiment No: 05

Date: 21/09/26

## Title: Map Interface Methods in Java Using HashMap

### 1. Objective / Aim

To study and demonstrate the commonly used methods of the Java Map interface using HashMap, including put(), get(), containsKey(), containsValue(), keySet(), values(), entrySet(), size(), isEmpty(), remove(), and clear().

### 2. System Requirements

#### Hardware Requirements

| Component    | Requirement                  |
|--------------|------------------------------|
| Processor    | Dual-core processor or above |
| RAM          | Minimum 4 GB                 |
| Storage      | At least 1 GB free space     |
| Input Device | Keyboard and mouse           |
| Display      | Standard monitor/display     |

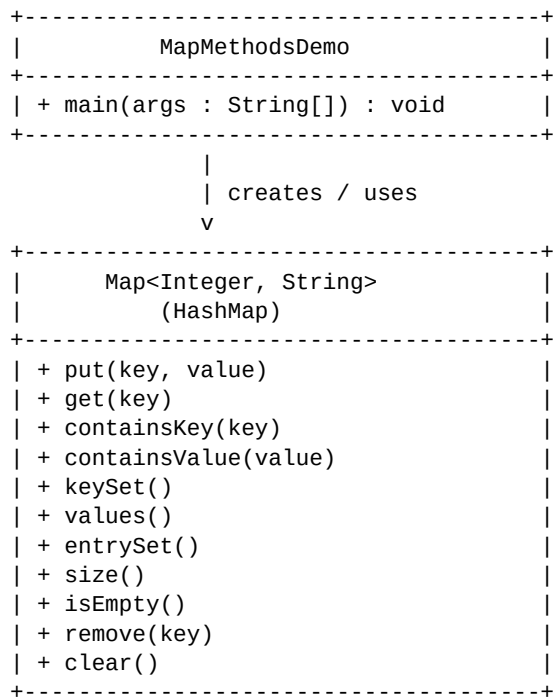
#### Software Requirements

| Software         | Requirement                                  |
|------------------|----------------------------------------------|
| Operating System | Windows / Linux / macOS                      |
| Java             | JDK 17 or JDK 21                             |
| IDE              | Eclipse / IntelliJ IDEA / NetBeans / VS Code |
| Terminal         | Command Prompt / PowerShell / Linux Terminal |

### 3. Algorithm / Logic Description

1. Start the program.
2. Import Map and HashMap from java.util.
3. Create a Map using HashMap.
4. Insert student key-value pairs using put().
5. Retrieve a value using get().
6. Check for a key and a value using containsKey() and containsValue().
7. Display keys, values, and key-value entries using keySet(), values(), and entrySet().
8. Display the number of mappings using size() and check whether the map is empty using isEmpty().
9. Remove key 104 using remove().
10. Remove all mappings using clear().
11. Check isEmpty() again after clear().
12. Stop the program.

## 4. UML Class Diagram



## 5. Source Code

```
import java.util.Map;
import java.util.HashMap;

class MapMethodsDemo {

    public static void main(String[] args) {

        // Creating a Map
        Map<Integer, String> students = new HashMap<>();

        // put()
        students.put(101, "Aarav");
        students.put(102, "Meera");
        students.put(103, "Rohan");
        students.put(104, "Diya");

        System.out.println("Map after put(): " + students);

        // get()
        System.out.println("Value for key 102: " + students.get(102));

        // containsKey()
        System.out.println("Contains key 103: "
            + students.containsKey(103));

        // containsValue()
        System.out.println("Contains value Rohan: "
            + students.containsValue("Rohan"));

        // keySet()
        System.out.println("Keys: " + students.keySet());

        // values()
        System.out.println("Values: " + students.values());

        // entrySet()
        System.out.println("Entries: " + students.entrySet());

        // size()
        System.out.println("Map size: " + students.size());

        // isEmpty()
        System.out.println("Is map empty: " + students.isEmpty());

        // remove()
        students.remove(104);
        System.out.println("After remove(104): " + students);

        // clear()
        students.clear();
        System.out.println("After clear(): " + students);

        // isEmpty() after clear
        System.out.println("Is map empty after clear: "
            + students.isEmpty());
    }
}
```

## 6. Compilation and Execution Commands

1. Save the program as:  
MapMethodsDemo.java
2. Open Command Prompt / Terminal in the folder containing the file.
3. Compile the program:  
javac MapMethodsDemo.java
4. Execute the program:

java MapMethodsDemo

## 7. Output

```
Map after put(): {101=Aarav, 102=Meera, 103=Rohan, 104=Diya}
Value for key 102: Meera
Contains key 103: true
Contains value Rohan: true
Keys: [101, 102, 103, 104]
Values: [Aarav, Meera, Rohan, Diya]
Entries: [101=Aarav, 102=Meera, 103=Rohan, 104=Diya]
Map size: 4
Is map empty: false
After remove(104): {101=Aarav, 102=Meera, 103=Rohan}
After clear(): {}
Is map empty after clear: true
```

## 8. Result / Completion

The program was successfully compiled and executed. The commonly used Map interface methods were demonstrated using a HashMap, and the expected output was obtained.